

5.8.20

EE25BTECH11022 - sankeerthan

Question: Five years ago, Minu was thrice as old as Sonu. Ten years later, Minu will be twice as old as Sonu. How old are Minu and Sonu?

Solution: Let Minu's present age be x and Sonu's present age be y . Five years ago, Minu was thrice as old as Sonu:

$$(1 \ 0) \begin{pmatrix} x \\ y \end{pmatrix} - 5 = 3 \left((0 \ 1) \begin{pmatrix} x \\ y \end{pmatrix} - 5 \right) \quad (1)$$

$$\left((1 \ 0) - 3(0 \ 1) \right) \begin{pmatrix} x \\ y \end{pmatrix} = -10 \quad (2)$$

$$(1 \ -3) \begin{pmatrix} x \\ y \end{pmatrix} = -10 \quad (3)$$

Ten years later, Minu will be twice as old as Sonu:

$$(1 \ 0) \begin{pmatrix} x \\ y \end{pmatrix} + 10 = 2 \left((0 \ 1) \begin{pmatrix} x \\ y \end{pmatrix} + 10 \right) \quad (4)$$

$$\left((1 \ 0) - 2(0 \ 1) \right) \begin{pmatrix} x \\ y \end{pmatrix} = 10 \quad (5)$$

$$(1 \ -2) \begin{pmatrix} x \\ y \end{pmatrix} = 10 \quad (6)$$

Matrix form:

$$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -10 \\ 10 \end{pmatrix} \quad (7)$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix}, \mathbf{B} = \begin{pmatrix} -10 \\ 10 \end{pmatrix} \quad (8)$$

Inverse of $\mathbf{A}$:

$$\mathbf{A}^{-1} = \begin{pmatrix} -2 & 3 \\ -1 & 1 \end{pmatrix} \quad (9)$$

$$\mathbf{X} = \mathbf{A}^{-1} \mathbf{B} \quad (10)$$

$$= \begin{pmatrix} -2 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -10 \\ 10 \end{pmatrix} \quad (11)$$

$$= \begin{pmatrix} 50 \\ 20 \end{pmatrix} \quad (12)$$

Therefore, Minu's present age = 50 and Sonu's present age = 20